

Course Projects

Final Project

- Groups of 3-4
- **Scope:** on the order of one of the programming assignments
- But, need to formulate the problem, design appropriate experiments, write up results in a formal report.

Selecting a Topic

- Part of your thesis? Great! (discuss with advisor)
- Find a problem you are interested in where you think NLP can help.
- OK to build on existing code / datasets.
- **First question:** what is the dataset?

Datasets

- Papers with code (<https://paperswithcode.com/area/natural-language-processing>)
- NLP Progress (<http://nlpprogress.com/>)
- SemEval Tasks:
 - <https://semeval.github.io/SemEval2023/tasks>
- Many more...

Requirements

- 1 Page Project Proposal (Group Formation + Project Topic)
 - Due 3/4/2025
- 4 Page Report
 - Due 4/25/2025
 - **Late project reports will not be accepted**
 - What is your contribution with respect to previous work?
 - Include empirical analysis of your approach
 - Report performance on dev / test set
 - Compare against appropriate baseline methods (example: Majority class + NBOW baseline)
- Code and data samples for the project (10MB max).
- A summary of the contributions made by each individual team member (1 page max).

Grading Rubric

For the reasonably well-prepared reader, is it clear what was done and why? Is the report well-written and well-structured?

Clarity (1-5):

How original is the approach? Does this project break new ground in topic, methodology, or content? How exciting and innovative is the work that it describes?

Originality / Innovativeness (1-5):

First, is the technical approach sound and well-chosen? Second, can one trust the claims of the report -- are they supported by proper experiments, proofs, or other argumentation?

Soundness / Correctness (1-5):

Does the author make clear where the problems and methods sit with respect to existing literature? Are any experimental results meaningfully compared with the best prior approaches?

Meaningful Comparison (1-5):

Overall (1-5):

General Advice

- **First question:** is the data you need easily available?
- Try to get a simple baseline working as early as possible to determine whether your project idea is feasible.
- Start with a manageable-sized dataset
 - Then scale up...

Suggested Project Directions

- Pick an existing dataset - identify the existing SOTA on the dataset. Analyze errors and hypothesize what techniques might help to address them. Implement, and compare to SOTA, and other sensible baselines.
- Pick a domain (e.g. scientific articles from some specific field), collect and annotate a dataset using BRAT, fine-tune some baseline models (e.g. BERT, T5, etc.).
- Formulate a question about some phenomenon that large language models (e.g. GPT-3 or LLaMA) can address (e.g. do they work well for numerical reasoning)?
 - Develop a new benchmark to test the capabilities on this task. How does your new benchmark relate to existing ones?
 - Evaluate several LLMs on your benchmark, and conduct analysis. What can we learn about the behavior of LLMs from this?

Group Formation Time (10 minutes)